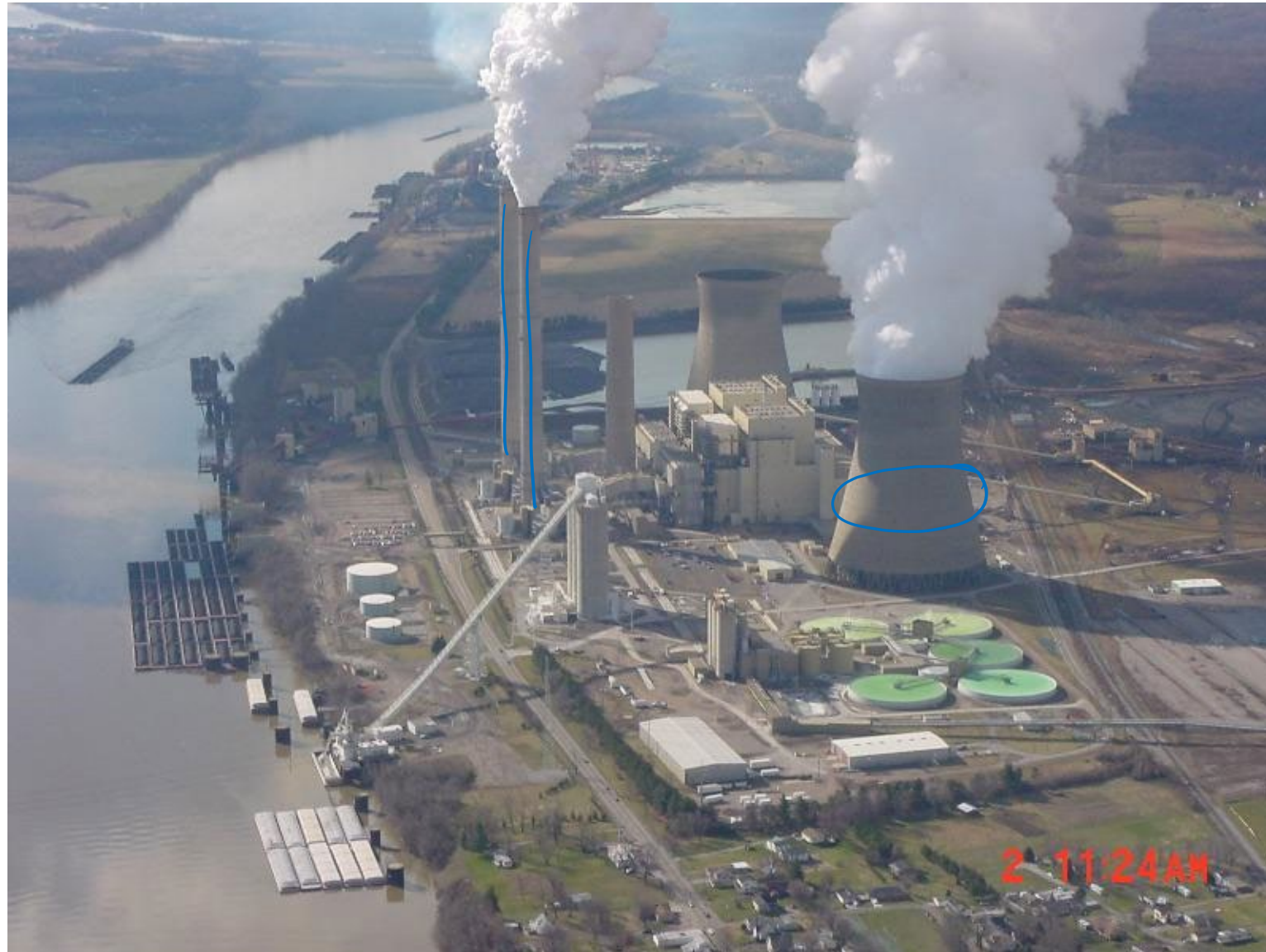




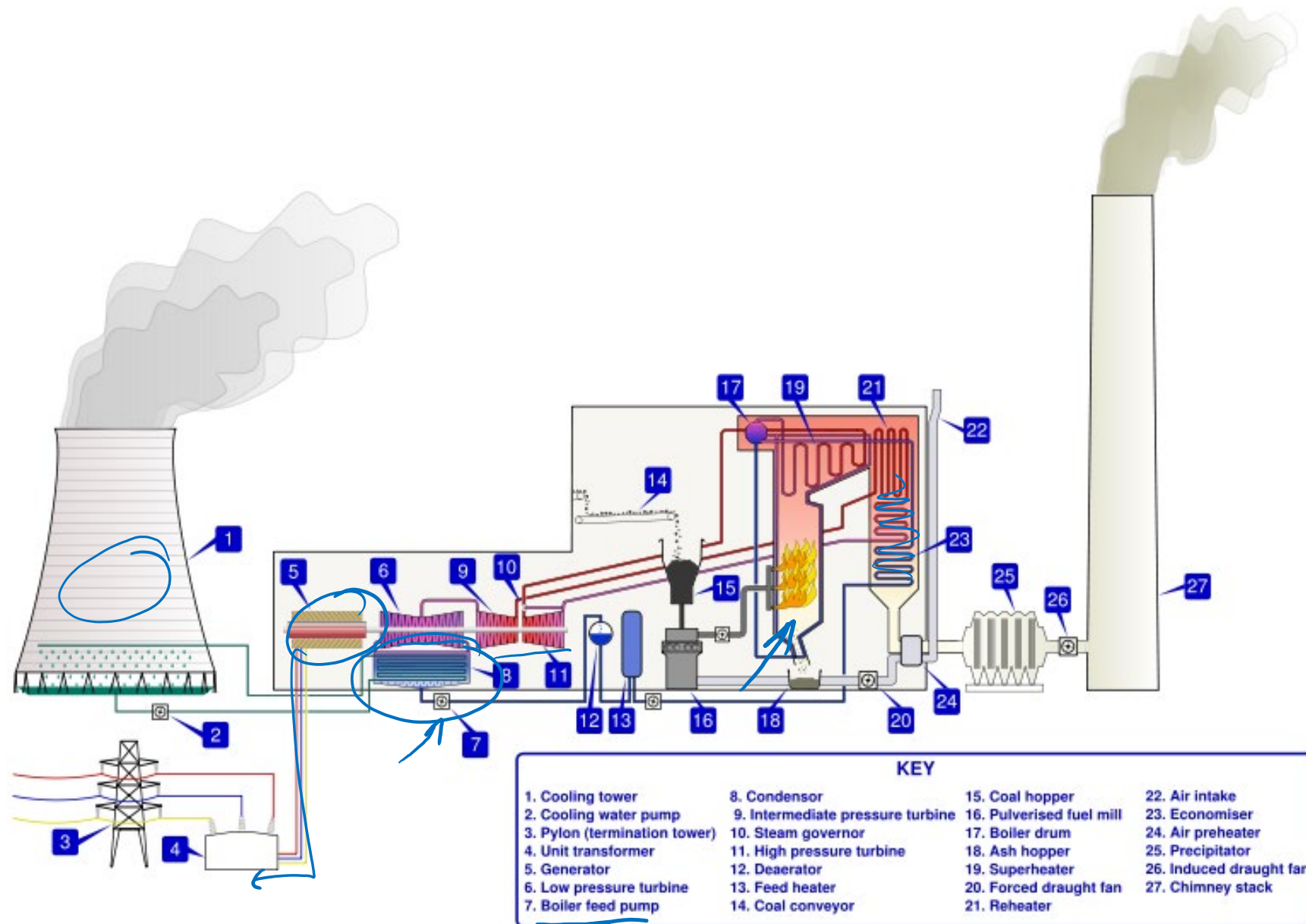
ME 300: Engineering Thermodynamics

Rankine Cycle

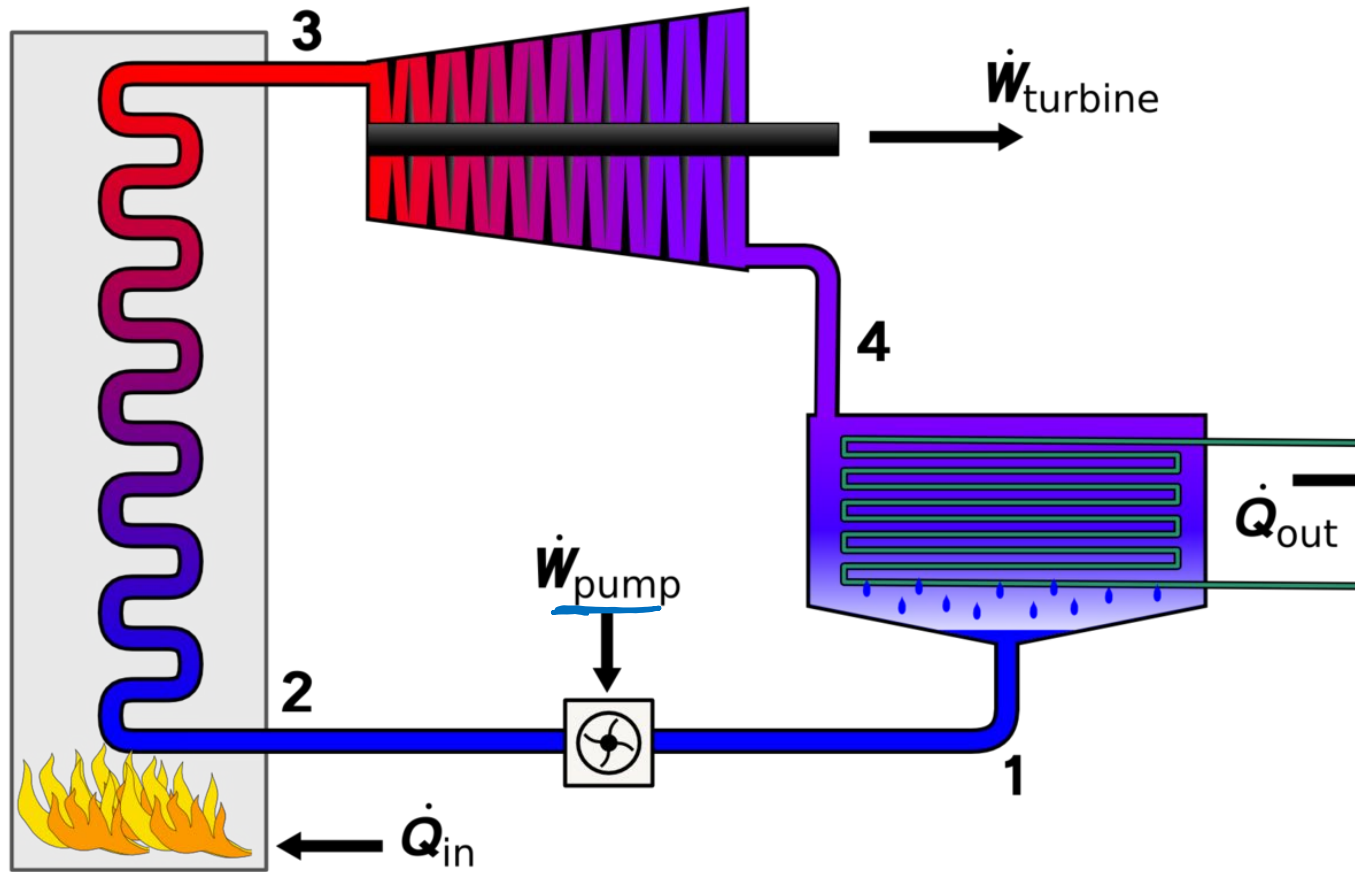
Steam power



Steam power



Rankine Cycle - ideal (reversible)



1-2 : isentropic compression
adiabatic, reversible

$\Rightarrow$ pump, liquid

2-3 : Isobaric heating

$\Rightarrow$ boiler, liq $\rightarrow$ vap

3-4 : isentropic expansion

$\Rightarrow$ turbine, vap $\rightarrow$ __

4-1 : isobaric heat rejection

$\Rightarrow$ condenser, __ $\rightarrow$ liq.

Rankine Cycle $\dot{m}(\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}$ steady-flow

$$1-2: \text{Pump} \Rightarrow \dot{m} \Delta h = -\dot{W}_2$$

$$2-3: \text{boiler} \Rightarrow \dot{m} \Delta h = \dot{Q}_3$$

$$3-4: \text{turbine} \Rightarrow \dot{m} \Delta h = -\dot{W}_4$$

$$4-1: \text{condenser} \Rightarrow \dot{m} \Delta h = \dot{Q}_1$$



What is the proper expression for the cycle efficiency?

A. $\frac{{}_3W_4}{{}_2Q_3}$

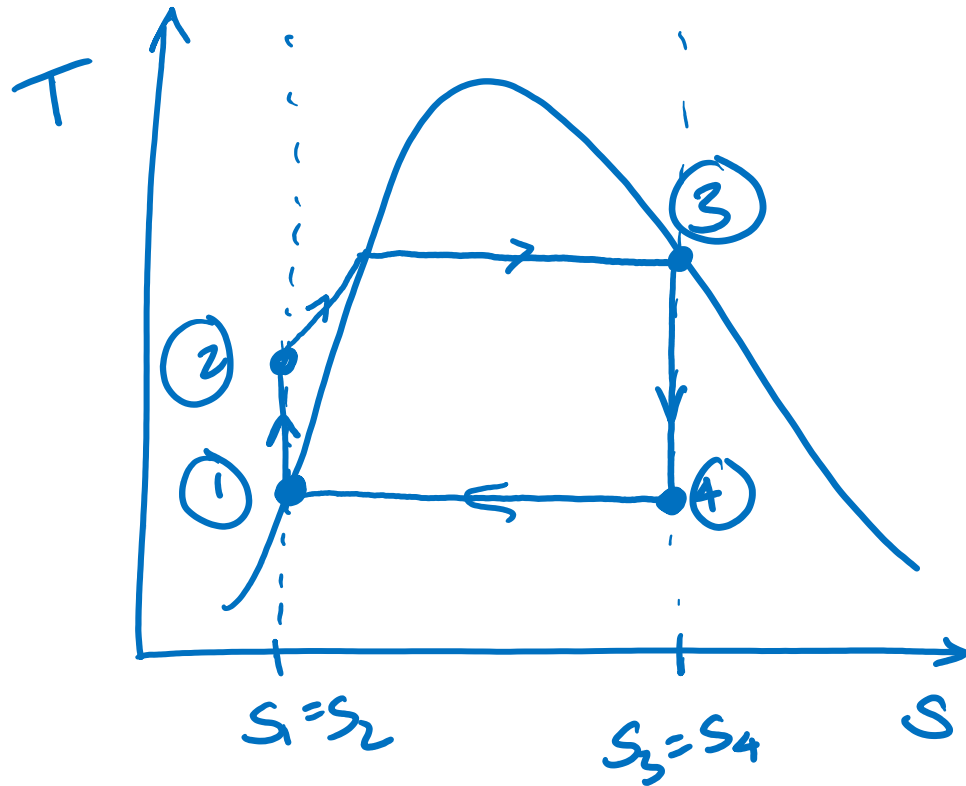
B. $\frac{{}_1W_2 + {}_3W_4}{{}_2Q_3}$

C. $\frac{{}_1W_2 + {}_3W_4}{{}_2Q_3 + {}_3Q_4}$

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_{in}}$$

Rankine Cycle

Normal - ideal



Superheated - ideal

